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Amrita Vishwavidyapeetham, Delhi NCR**

Project Report

Computational Analysis and Optimization of Quantum Harmonic Oscillator Wavefunctions

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Abstract

This report presents a detailed computational analysis of the quantum harmonic oscillator, a fundamental model in quantum mechanics. The study employs Hermite polynomials and dimensionless coordinate transformations to accurately compute the wavefunctions of the first five energy states. These wavefunctions, along with their corresponding probability densities and energy eigenvalues, are graphically visualized to illustrate their spatial behavior and quantum characteristics. Orthogonality between the wavefunctions is numerically verified through overlap integrals, ensuring the physical validity of the solutions. Additionally, the superposition of states is analyzed, comparing the original and optimized combinations to highlight the impact on wavefunction localization and probability densities. The probability of locating the quantum particle within a confined region around the equilibrium position is also quantitatively assessed. This approach demonstrates the effectiveness of combining analytical formulations with computational visualization for a deeper understanding of quantum behavior. The results have significant implications for real-world quantum systems, such as vibrational modes in molecules or confined electrons in quantum dots. By improving insight into wavefunction shapes and quantized energy levels, the study contributes to the design and development of nanoscale devices and emerging quantum technologies, bridging theoretical principles with practical applications.

1 Objectives of the Project

- **To computationally model the quantum harmonic oscillator system** by solving the Schrödinger equation for different quantized energy levels.
- **To optimize the oscillator frequency by minimizing quantum superposition(ω)** by minimizing the expected ground state energy, thereby identifying the most stable quantum state.
- **To calculate and visualize the wavefunctions and probability densities** of the first five energy eigenstates using Hermite polynomials and dimensionless coordinates.
- **To verify the orthogonality of the computed wavefunctions**, confirming that each quantum state is physically distinct and independent.
- **To analyze the probability distribution of the particle's position**, especially the likelihood of finding it within specific spatial regions.
- **To explore the relationship between potential energy, wavefunctions, and discrete energy levels** inherent to the quantum harmonic oscillator.
- **To demonstrate the practical significance of the quantum harmonic oscillator model** in describing real-world quantum systems such as molecular vibrations and electron confinement in quantum dots.

2 Literature Survey

The quantum harmonic oscillator is a cornerstone model in quantum mechanics, representing a particle subject to a quadratic potential well. Its analytical solution via the Schrödinger equation provides discrete energy eigenvalues and corresponding wavefunctions characterized by Hermite polynomials [6]. This model not only offers fundamental insights into quantum behavior but also serves as an approximation for various physical systems, including molecular vibrations and phonons in solids [7].

The time-independent Schrödinger equation governs the quantum states of a system, and its solutions reveal quantized energy levels and probability distributions for particles [8]. Analytical solutions exist for simple potentials like the harmonic oscillator; however, more complex or optimized parameters require numerical methods.

Optimization techniques, such as constrained minimization, have been applied in quantum mechanics to find system parameters that minimize energy expectation values, leading to more accurate and physically meaningful solutions [9]. These methods are particularly useful when analytical solutions are difficult or when adjusting parameters to fit experimental data.

MATLAB provides a versatile computational environment to implement these numerical methods efficiently. It allows for symbolic and numeric solutions of differential equations, plotting of wavefunctions, and execution of optimization algorithms like `fmincon` [10]. MATLAB's capabilities facilitate the simulation and visualization of quantum systems, enabling deeper understanding through computational experiments [11].

This project builds on these foundations by integrating optimization with quantum harmonic oscillator modeling, demonstrating the synergy between theoretical physics and numerical computing.

3 Introduction to Quantum Mechanics for Harmonic Oscillator Project

Quantum mechanics is a fundamental framework in modern physics that describes the behavior of matter and energy on microscopic scales, such as atoms and subatomic particles. Unlike classical mechanics, which predicts definite trajectories for particles, quantum mechanics employs probabilistic descriptions of physical systems. This introduction outlines the key concepts of quantum mechanics relevant to the study of a one-dimensional quantum harmonic oscillator, as implemented in the provided MATLAB code, where the potential is given by $V(x) = \alpha x^2$ with $\alpha = \frac{1}{2}m\omega^2$. These concepts include Hilbert spaces, dual vectors, Hermitian operators, Hermite polynomials, the Schrödinger equation, time evolution, wave packets, and energy quantization.

3.1 Hilbert Space

In quantum mechanics, the state of a system is represented by a wavefunction $\psi(x)$, which resides in a complex vector space known as a **Hilbert space**. A Hilbert space is a complete, infinite-dimensional vector space equipped with an inner product, allow-

ing for the computation of probabilities and expectation values. For the quantum harmonic oscillator, the Hilbert space consists of square-integrable functions $L^2(\mathbb{R})$, meaning $\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty$. This ensures that the wavefunction is normalizable, a requirement for physical states to yield finite probabilities. In the provided code, the wavefunctions $\psi_n(x)$ for the first five energy states are computed and stored in a matrix, with their orthogonality verified in the Hilbert space via the overlap integral $\langle \psi_n | \psi_m \rangle = \delta_{nm}$.

3.2 Dual Vectors and Bra-Ket Notation

Quantum states are often expressed using Dirac's **bra-ket notation**, where a state vector is denoted as a “ket” $|\psi\rangle$, and its conjugate transpose, or “bra,” is $\langle\psi|$. The inner product between two states $|\psi\rangle$ and $|\phi\rangle$ is written as:

$$\langle\phi|\psi\rangle = \int_{-\infty}^{\infty} \phi^*(x)\psi(x) dx$$

The dual vector $\langle\psi|$ is the Hermitian conjugate of $|\psi\rangle$. In the code, the orthogonality check computes integrals like $\langle\psi_i|\psi_j\rangle$, forming an orthogonality matrix that approximates the identity matrix, confirming that the wavefunctions form an orthonormal basis in the Hilbert space.

3.3 Hermitian Operators and Hermitian Conjugate

Operators in quantum mechanics, such as position $\hat{x}$, momentum $\hat{p} = -i\hbar\frac{d}{dx}$, and the Hamiltonian $\hat{H}$, act on state vectors to produce observables. An operator $\hat{A}$ is **Hermitian** if it equals its Hermitian conjugate, $\hat{A}^\dagger = \hat{A}$, ensuring real eigenvalues for physical observables. The Hermitian conjugate of an operator is defined such that:

$$\langle\phi|\hat{A}|\psi\rangle = \langle\psi|\hat{A}^\dagger|\phi\rangle^*$$

For the harmonic oscillator, the Hamiltonian is:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2$$

This operator is Hermitian, guaranteeing real energy eigenvalues. The code approximates the kinetic energy term using a finite difference method for the second derivative to compute the ground state energy expectation value.

3.4 Hermite Polynomials

The eigenfunctions of the quantum harmonic oscillator are expressed using **Hermite polynomials**, which arise in the solution of the time-independent Schrödinger equation. For the potential $V(x) = \frac{1}{2}m\omega^2 x^2$, the wavefunctions are:

$$\psi_n(x) = N_n H_n(\xi) e^{-\xi^2/2}, \quad \xi = \sqrt{\frac{m\omega}{\hbar}} x$$

where $H_n(\xi)$ is the n -th Hermite polynomial (physicists' version), and $N_n = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}}$ is the normalization constant. In the code, the MATLAB function `hermiteH` computes these polynomials, and the wavefunctions for $n = 1$ to 5 are calculated and plotted, with normalization verified via $\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1$.

3.5 Schrödinger Equation

The **Schrödinger equation** governs the behavior of quantum systems. The time-independent form, used for stationary states, is:

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$$

For the harmonic oscillator, this yields quantized energy levels:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

The code computes these energy levels using the optimized ω_{opt} and plots them alongside the potential $V(x)$. The time-dependent Schrödinger equation, $i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$, describes the evolution of non-stationary states but is not directly implemented in the code, as the focus is on stationary states.

3.6 Energy Quantization

Unlike classical mechanics, where energy can take any value, quantum mechanics imposes **energy quantization**. For the harmonic oscillator, the energy levels are discrete, as given by $E_n = \hbar\omega(n + \frac{1}{2})$. The code computes these energies for $n = 1$ to 5 using the optimized ω_{opt} , and the results are visualized in plots alongside the potential, highlighting the quantized nature of the system.

3.7 Relevance to the Project

The project analyzes a quantum harmonic oscillator with the potential $V(x) = \frac{1}{2}m\omega^2x^2$, where ω is optimized to minimize the ground state energy using the `fmincon` function in MATLAB. The wavefunctions are computed using Hermite polynomials, and their properties (normalization, orthogonality, and probability distributions) are verified numerically. The position range $x \in [-4\text{ nm}, 4\text{ nm}]$ and electron mass $m = 9.1 \times 10^{-31}\text{ kg}$ align with nanoscale systems, such as quantum dots or molecular vibrations. This quantum mechanical framework provides insights into the behavior of such systems, contrasting with the classical harmonic oscillator's continuous energy and definite trajectories.

4 Classical Description of the Quantum Harmonic Oscillator System

The physical system under consideration is a one-dimensional quantum harmonic oscillator, which can be described classically to provide a foundational understanding before delving into its quantum mechanical treatment. In the classical framework, the harmonic oscillator consists of a particle of mass m (in this case, an electron with mass $m = 9.1 \times 10^{-31}\text{ kg}$) subjected to a restoring force proportional to its displacement from an equilibrium position. This force is described by Hooke's law,

$$F = -kx,$$

where k is the spring constant, and x is the displacement from equilibrium.

The potential energy of the system is given by:

$$V(x) = \frac{1}{2}kx^2.$$

The angular frequency ω of the oscillator is related to the spring constant and mass by:

$$\omega = \sqrt{\frac{k}{m}}.$$

Thus, the potential energy can be rewritten as:

$$V(x) = \frac{1}{2}m\omega^2x^2.$$

In this classical system, the particle oscillates sinusoidally about the equilibrium position ($x = 0$) with a frequency determined by ω . The total energy E of the system is the sum of kinetic energy T and potential energy V :

$$E = T + V = \frac{1}{2}mv^2 + \frac{1}{2}m\omega^2x^2,$$

This classical description provides a useful contrast to the quantum mechanical treatment, where the particle's position is described by a probability distribution (wavefunction) rather than a definite trajectory, and energy is quantized into discrete levels. The classical harmonic oscillator serves as a reference for understanding the transition to quantum behavior, particularly in visualizing the potential energy and the spatial extent of the particle's motion.

5 Quantum Realization of the Quantum Harmonic Oscillator

This section analytically solves the time-independent Schrödinger equation for the quantum harmonic oscillator (QHO) with potential $V(x) = \alpha x^2$, where $\alpha = \frac{1}{2}m\omega^2$. General expressions are derived for the wavefunctions, probability density, energy eigenvalues, normalization constant, orthogonality condition, expectation values, and the cost function used to optimize the angular frequency ω .

5.1 Time-Independent Schrödinger Equation

The time-independent Schrödinger equation is:

$$\hat{H}\psi_n(x) = E_n\psi_n(x) \tag{1}$$

The Hamiltonian for the QHO with potential $V(x) = \frac{1}{2}m\omega^2x^2$ is:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2 \tag{2}$$

Thus, the equation becomes:

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2\right) \psi_n(x) = E_n \psi_n(x) \quad (3)$$

where $\hbar$ is the reduced Planck constant, m is the particle mass, ω is the angular frequency, $\psi_n(x)$ is the wavefunction, and E_n is the energy eigenvalue. The potential coefficient is $\alpha = \frac{1}{2}m\omega^2$.

5.2 Solving the Schrödinger Equation

Introduce a dimensionless coordinate:

$$\beta = \frac{m\omega}{\hbar}, \quad \xi = \sqrt{\beta}x = \sqrt{\frac{m\omega}{\hbar}}x \quad (4)$$

Transform the derivatives:

$$x = \xi \sqrt{\frac{\hbar}{m\omega}}, \quad \frac{d}{dx} = \sqrt{\beta} \frac{d}{d\xi}, \quad \frac{d^2}{dx^2} = \beta \frac{d^2}{d\xi^2} \quad (5)$$

The potential term is:

$$\frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2 \frac{\xi^2}{\beta} = \frac{1}{2}\hbar\omega \xi^2 \quad (6)$$

Substitute into the Schrödinger equation:

$$-\frac{\hbar^2}{2m} \beta \frac{d^2 \psi_n}{d\xi^2} + \frac{1}{2}\hbar\omega \xi^2 \psi_n = E_n \psi_n \quad (7)$$

Since $\frac{\hbar^2}{2m} \beta = \frac{\hbar\omega}{2}$, this becomes:

$$-\frac{\hbar\omega}{2} \frac{d^2 \psi_n}{d\xi^2} + \frac{1}{2}\hbar\omega \xi^2 \psi_n = E_n \psi_n \quad (8)$$

Divide by $\frac{1}{2}\hbar\omega$:

$$\frac{d^2 \psi_n}{d\xi^2} - \xi^2 \psi_n = -\frac{2E_n}{\hbar\omega} \psi_n \quad (9)$$

5.3 Asymptotic Behavior and Trial Solution

For large $|\xi|$, the equation approximates to:

$$\frac{d^2\psi_n}{d\xi^2} \approx \xi^2\psi_n \quad (10)$$

Try $\psi_n(\xi) = e^{-\xi^2/2}$:

$$\frac{d\psi_n}{d\xi} = -\xi e^{-\xi^2/2}, \quad \frac{d^2\psi_n}{d\xi^2} = (\xi^2 - 1)e^{-\xi^2/2} \quad (11)$$

This suggests a Gaussian factor. Assume:

$$\psi_n(\xi) = H(\xi)e^{-\xi^2/2} \quad (12)$$

Compute derivatives:

$$\frac{d\psi_n}{d\xi} = \frac{dH}{d\xi}e^{-\xi^2/2} - \xi H e^{-\xi^2/2} \quad (13)$$

$$\frac{d^2\psi_n}{d\xi^2} = \left(\frac{d^2H}{d\xi^2} - 2\xi \frac{dH}{d\xi} + (\xi^2 - 1)H \right) e^{-\xi^2/2} \quad (14)$$

Substitute:

$$\left(\frac{d^2H}{d\xi^2} - 2\xi \frac{dH}{d\xi} - H + \frac{2E_n}{\hbar\omega} H \right) e^{-\xi^2/2} = 0 \quad (15)$$

This yields the Hermite differential equation:

$$\frac{d^2H}{d\xi^2} - 2\xi \frac{dH}{d\xi} + \left(\frac{2E_n}{\hbar\omega} - 1 \right) H = 0 \quad (16)$$

Solutions exist when:

$$\frac{2E_n}{\hbar\omega} - 1 = 2n, \quad E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (17)$$

The solutions are the Hermite polynomials $H_n(\xi)$ (physicists' version).

5.4 Wavefunctions

The wavefunctions are:

$$\psi_n(x) = N_n H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega}{2\hbar} x^2} \quad (18)$$

The first five Hermite polynomials are:

$$\begin{aligned} H_0(\xi) &= 1 \\ H_1(\xi) &= 2\xi \\ H_2(\xi) &= 4\xi^2 - 2 \\ H_3(\xi) &= 8\xi^3 - 12\xi \\ H_4(\xi) &= 16\xi^4 - 48\xi^2 + 12 \end{aligned}$$

Thus:

$$\psi_0(x) = N_0 e^{-\frac{m\omega}{2\hbar} x^2} \quad (19)$$

$$\psi_1(x) = N_1 \cdot 2\sqrt{\frac{m\omega}{\hbar}} x e^{-\frac{m\omega}{2\hbar} x^2} \quad (20)$$

$$\psi_2(x) = N_2 \left(4\frac{m\omega}{\hbar} x^2 - 2 \right) e^{-\frac{m\omega}{2\hbar} x^2} \quad (21)$$

$$\psi_3(x) = N_3 \left(8\left(\frac{m\omega}{\hbar}\right)^{3/2} x^3 - 12\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega}{2\hbar} x^2} \quad (22)$$

$$\psi_4(x) = N_4 \left(16\left(\frac{m\omega}{\hbar}\right)^2 x^4 - 48\frac{m\omega}{\hbar} x^2 + 12 \right) e^{-\frac{m\omega}{2\hbar} x^2} \quad (23)$$

5.5 Normalization

The normalization condition is:

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1 \quad (24)$$

$$|\psi_n(x)|^2 = N_n^2 [H_n(\xi)]^2 e^{-\xi^2}, \quad \xi = \sqrt{\frac{m\omega}{\hbar}} x$$

Using $dx = \sqrt{\frac{\hbar}{m\omega}} d\xi$:

$$N_n^2 \sqrt{\frac{\hbar}{m\omega}} \int_{-\infty}^{\infty} [H_n(\xi)]^2 e^{-\xi^2} d\xi = 1 \quad (25)$$

$$\int_{-\infty}^{\infty} [H_n(\xi)]^2 e^{-\xi^2} d\xi = \sqrt{\pi} 2^n n!$$

$$N_n^2 = \frac{1}{\sqrt{\pi} 2^n n! \sqrt{\frac{\hbar}{m\omega}}}, \quad N_n = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}}$$

5.6 Probability Density

The probability density is:

$$|\psi_n(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \frac{1}{2^n n!} [H_n(\xi)]^2 e^{-\xi^2} \quad (26)$$

For example:

$$|\psi_0(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega}{\hbar}x^2} \quad (27)$$

$$|\psi_1(x)|^2 = 2 \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \frac{m\omega}{\hbar} x^2 e^{-\frac{m\omega}{\hbar}x^2} \quad (28)$$

5.7 Energy Eigenvalues

The energy eigenvalues are:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, 3, 4 \quad (29)$$

5.8 Orthogonality

The wavefunctions are orthogonal:

$$\langle \psi_n | \psi_m \rangle = \int_{-\infty}^{\infty} \psi_n^*(x) \psi_m(x) dx = \delta_{nm} \quad (30)$$

$$\langle \psi_n | \psi_m \rangle = N_n N_m \sqrt{\frac{\hbar}{m\omega}} \sqrt{\pi} 2^n n! \delta_{nm} = \delta_{nm}$$

5.9 Position Variance and Regularization in Superposed Wavefunctions

`position_variance_superposition(w, psi_set, x, dx)`

This function computes the **position variance** of a superposed wavefunction:

$$\psi(x) = \sum_n w_n \psi_n(x)$$

Where: - $\psi_n(x)$ are basis eigenstates, - w_n are their corresponding weights.

The function evaluates the expected position and squared position:

$$\langle x \rangle = \int x |\psi(x)|^2 dx, \quad \langle x^2 \rangle = \int x^2 |\psi(x)|^2 dx$$

Then it computes the position variance:

$$\text{Var}(x) = \langle x^2 \rangle - \langle x \rangle^2$$

This variance reflects how spatially localized or delocalized the resulting wavefunction is. The primary optimization goal is to **minimize this spatial spread**, favoring more localized quantum states.

Regularization Term: $\lambda \sum_n n^2 |w_n|^2$

This term acts as a **regularizer** to discourage the use of high-energy basis states:

- $|w_n|^2$: probability amplitude for each basis state,
- n^2 : penalizes higher energy contributions (since QHO energy increases with $n+1/2$).

The full regularization penalty is:

$$\text{Penalty} = \lambda \sum_n n^2 |w_n|^2$$

- **Purpose:** Penalizes contributions from high n , i.e., more oscillatory or energy-intensive components.
- **Effect:** Encourages smoother and more physically plausible solutions.
- λ : A tunable hyperparameter that controls the strength of this penalty.

Final Cost Function:

$$\text{Total Cost} = \underbrace{\text{Var}(x)}_{\text{position spread}} + \underbrace{\lambda \sum_n n^2 |w_n|^2}_{\text{energy regularization}}$$

This cost function balances two objectives:

- **Minimizing spatial spread** of the wavefunction in position space.
- **Restricting energy usage** by penalizing high-energy quantum state contributions.

5.10 Time-Independent Schrödinger Equation

The time-independent Schrödinger equation is given by:

$$\hat{H}\psi_n(x) = E_n\psi_n(x), \tag{31}$$

where $\hat{H}$ is the Hamiltonian, $\psi_n(x)$ is the wavefunction for the n -th energy state, and E_n is the corresponding energy eigenvalue. For the QHO with potential $V(x) = \frac{1}{2}m\omega^2x^2$, the Hamiltonian is:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2. \quad (32)$$

Thus, the Schrödinger equation becomes:

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2 \right) \psi_n(x) = E_n \psi_n(x). \quad (33)$$

5.11 Parameters

The given parameters are:

- Reduced Planck constant: $\hbar = 1.0545718e - 34$ J·s
- Electron mass: $m = 9.1e - 31$ kg
- Angular frequency: $\omega = 6.283e14$ rad/s
- Position range: $x \in [-4e - 9, 4e - 9]$ m
- Number of points: $N = 2000$

5.12 Potential

The potential is $V(x) = \alpha x^2$, with $\alpha = \frac{1}{2}m\omega^2$. Calculating α :

$$\begin{aligned} \omega^2 &= (6.283e14)^2 = 3.947689e29 \text{ rad}^2/\text{s}^2, \\ \frac{1}{2}m &= \frac{1}{2} \times 9.1e - 31 = 4.55e - 31 \text{ kg}, \\ \alpha &= 4.55e - 31 \times 3.947689e29 = 0.1796198 \text{ N/m}. \end{aligned}$$

Thus, $V(x) = 0.1796198x^2$.

5.13 Energy Eigenvalues

The energy eigenvalues for the QHO are:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, 3, 4, \dots \quad (34)$$

Calculate $\hbar\omega$:

$$\hbar\omega = 1.0545718e - 34 \times 6.283e14 = 6.625948e - 20 \text{ J}.$$

For $n = 0$ to 4:

$$\begin{aligned}
E_0 &= \frac{1}{2} \times 6.625948e - 20 = 3.312974e - 20 \text{ J}, \\
E_1 &= \frac{3}{2} \times 6.625948e - 20 = 9.938844e - 20 \text{ J}, \\
E_2 &= \frac{5}{2} \times 6.625948e - 20 = 1.656487e - 19 \text{ J}, \\
E_3 &= \frac{7}{2} \times 6.625948e - 20 = 2.319379e - 19 \text{ J}, \\
E_4 &= \frac{9}{2} \times 6.625948e - 20 = 2.981676e - 19 \text{ J}.
\end{aligned}$$

5.14 Wavefunctions

The wavefunctions are:

$$\psi_n(x) = N_n H_n(\xi) e^{-\xi^2/2}, \quad \xi = \sqrt{\frac{m\omega}{\hbar}} x, \quad (35)$$

where $H_n(\xi)$ is the n -th Hermite polynomial, and the normalization constant is:

$$N_n = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \frac{1}{\sqrt{2^n n!}}. \quad (36)$$

Calculate the scaling factor:

$$\begin{aligned}
\frac{m\omega}{\hbar} &= \frac{9.1e - 31 \times 6.283e14}{1.0545718e - 34} = 5.417628e18 \text{ m}^{-2}, \\
\sqrt{\frac{m\omega}{\hbar}} &= \sqrt{5.417628e18} = 2.327689e9 \text{ m}^{-1}.
\end{aligned}$$

So, $\xi = 2.327689e9x$. For the normalization constant:

$$\begin{aligned}
\frac{m\omega}{\pi\hbar} &= \frac{9.1e - 31 \times 6.283e14}{3.14159265359 \times 1.0545718e - 34} = 1.724195e18 \text{ m}^{-2}, \\
\left(\frac{m\omega}{\pi\hbar} \right)^{1/4} &\approx 36244.69 \text{ m}^{-1/2}.
\end{aligned}$$

The normalization constants are:

$$\begin{aligned}
N_0 &= 36244.69 \text{ m}^{-1/2}, \\
N_1 &= \frac{36244.69}{\sqrt{2}} \approx 25629.39 \text{ m}^{-1/2}, \\
N_2 &= \frac{36244.69}{\sqrt{8}} \approx 12814.69 \text{ m}^{-1/2}, \\
N_3 &= \frac{36244.69}{\sqrt{48}} \approx 5230.84 \text{ m}^{-1/2}, \\
N_4 &= \frac{36244.69}{\sqrt{384}} \approx 1849.88 \text{ m}^{-1/2}.
\end{aligned}$$

5.15 Hermite Polynomials

The first five Hermite polynomials (physicists' version) are:

$$\begin{aligned}H_0(\xi) &= 1, \\H_1(\xi) &= 2\xi, \\H_2(\xi) &= 4\xi^2 - 2, \\H_3(\xi) &= 8\xi^3 - 12\xi, \\H_4(\xi) &= 16\xi^4 - 48\xi^2 + 12.\end{aligned}$$

5.16 Numerical Computation

The position space is discretized over $x = [-4e - 9, 4e - 9]$ m with $N = 2000$ points:

$$dx = \frac{8e - 9}{1999} \approx 4.002001e - 12 \text{ m}.$$

The wavefunction is computed as:

$$\psi_n(x) = N_n H_n(2.327689e9x) e^{-2.708814e18x^2}.$$

5.17 Sample Calculations

At $x = 0$:

$$\begin{aligned}\psi_0(0) &= 36244.69 \text{ m}^{-1/2}, \\ \psi_1(0) &= 0, \\ \psi_2(0) &= -25629.38 \text{ m}^{-1/2}, \\ \psi_3(0) &= 0, \\ \psi_4(0) &= 22198.56 \text{ m}^{-1/2}.\end{aligned}$$

At $x = 1e - 9$ m:

$$\begin{aligned}\psi_0 &= 2415.05 \text{ m}^{-1/2}, \\ \psi_1 &= 7952.64 \text{ m}^{-1/2}, \\ \psi_2 &= 16799.76 \text{ m}^{-1/2}, \\ \psi_3 &= 25429.28 \text{ m}^{-1/2}, \\ \psi_4 &= 27267.48 \text{ m}^{-1/2}.\end{aligned}$$

5.18 Normalization and Orthogonality

The wavefunctions are normalized:

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1. \quad (37)$$

Numerically approximated using the trapezoidal rule over the finite range. The wavefunctions are orthogonal:

$$\langle \psi_n | \psi_m \rangle = \delta_{nm}. \quad (38)$$

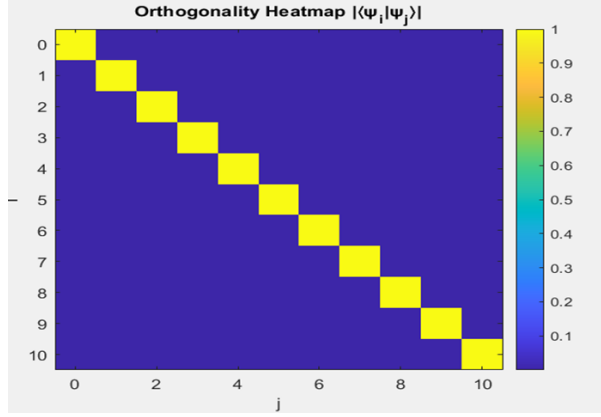


Figure 1: orthogonal overlap matrix heatmap

6 Simulation and Analysis Results

This section presents the simulation results for the quantum harmonic oscillator (QHO) with potential $V(x) = \frac{1}{2}m\omega^2x^2$, analyzing the wavefunctions $\psi_n(x)$ for $n = 1$ to 5, their probability densities, energy levels, and the optimization of the angular frequency ω . The simulation includes numerical computations over the position range $x \in [-4 \text{ nm}, 4 \text{ nm}]$, using 2000 discrete points.

We construct the superposed wavefunction as:

$$\psi(x) = \sum_n w_n \psi_n(x)$$

where $\psi_n(x)$ are harmonic oscillator eigenstates and w_n are the learned coefficients.

6.1 Position Variance

The expectation values are computed as:

$$\langle x \rangle = \int x |\psi(x)|^2 dx, \quad \langle x^2 \rangle = \int x^2 |\psi(x)|^2 dx$$

The position variance is given by:

$$\text{Var}(x) = \langle x^2 \rangle - \langle x \rangle^2$$

This variance indicates the spatial spread of the wavefunction, which the optimizer seeks to minimize.

6.2 Regularization

To avoid high-energy contributions, we apply a regularization term:

$$\lambda \sum_n n^2 |w_n|^2$$

This penalizes higher quantum numbers and encourages smoother, lower-energy solutions. The hyperparameter λ controls the strength of this penalty.

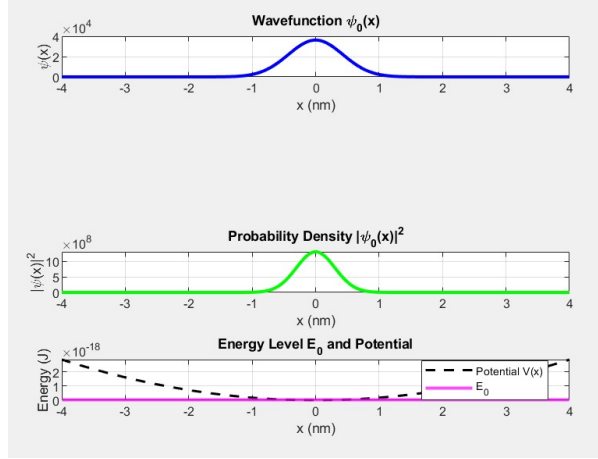


Figure 2: $n=0$

6.3 Total Cost Function

The overall cost function minimized during optimization is:

$$\text{Cost} = \text{Var}(x) + \lambda \sum_n n^2 |w_n|^2$$

This balances spatial localization with energy efficiency in the quantum superposition.

6.4 Wavefunctions and Probability Densities

The wavefunctions $\psi_n(x)$ for $n = 1$ to 5 were computed using the analytical form:

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n \left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega}{2\hbar} x^2}$$

Figures ?? to ?? show the wavefunctions (blue), probability densities $|\psi_n(x)|^2$ (green), and energy levels E_n (pink dashed line) alongside the potential $V(x)$ (black dashed line) for each state.

Figure ?? shows all wavefunctions ψ_1 to ψ_5 overlaid, illustrating the transformation from the first to the fifth state. The number of nodes increases with n , from 0 ($n = 1$) to 4 ($n = 5$), consistent with the QHO's quantum behavior.

6.5 Normalization and Orthogonality

The total probability for each state was computed numerically over the finite range:

$$\int_{-4\text{ nm}}^{4\text{ nm}} |\psi_n(x)|^2 dx \approx 1$$

The results are 1.0000 for all states ($n = 1$ to 5), confirming proper normalization within numerical precision. A discrete sum estimate for ψ_1 yields 0.99950, slightly less than 1 due to the finite integration range.

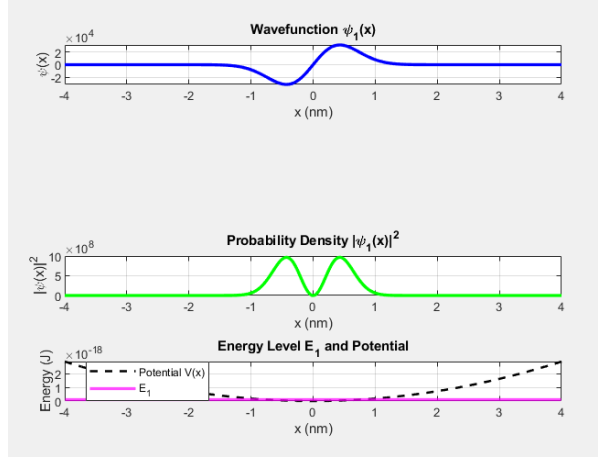


Figure 3: Wavefunction $\psi_1(x)$, probability density $|\psi_1(x)|^2$, energy level E_1 , and potential $V(x)$.

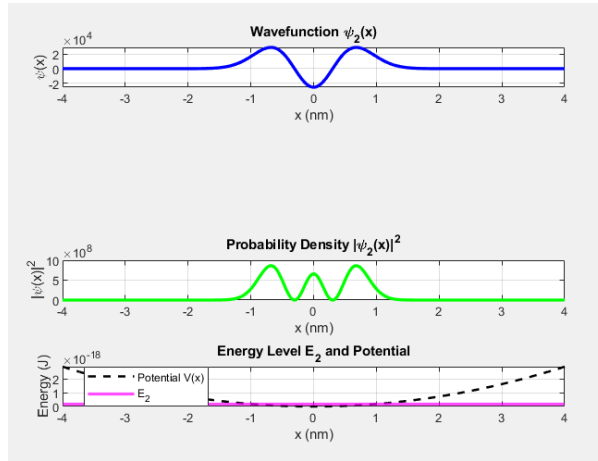


Figure 4: Wavefunction $\psi_2(x)$, probability density $|\psi_2(x)|^2$, energy level E_2 , and potential $V(x)$.

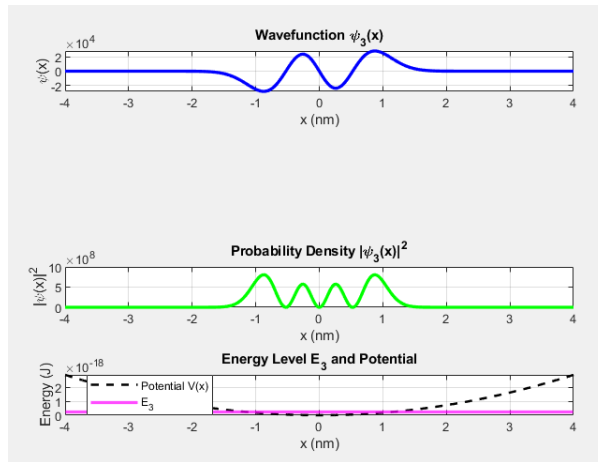


Figure 5: Wavefunction $\psi_3(x)$, probability density $|\psi_3(x)|^2$, energy level E_3 , and potential $V(x)$.

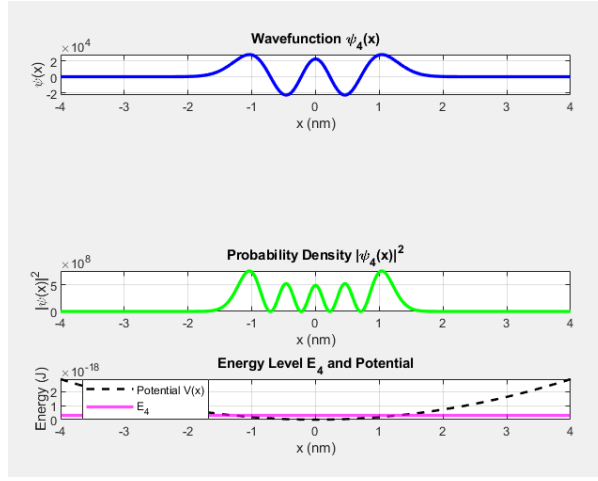


Figure 6: Wavefunction $\psi_4(x)$, probability density $|\psi_4(x)|^2$, energy level E_4 , and potential $V(x)$.

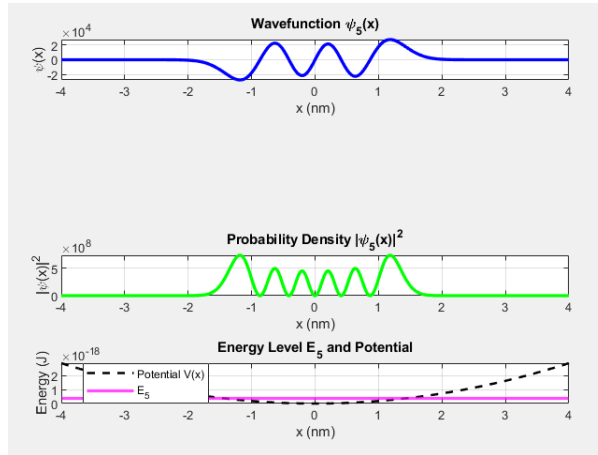


Figure 7: Wavefunction $\psi_5(x)$, probability density $|\psi_5(x)|^2$, energy level E_5 , and potential $V(x)$.

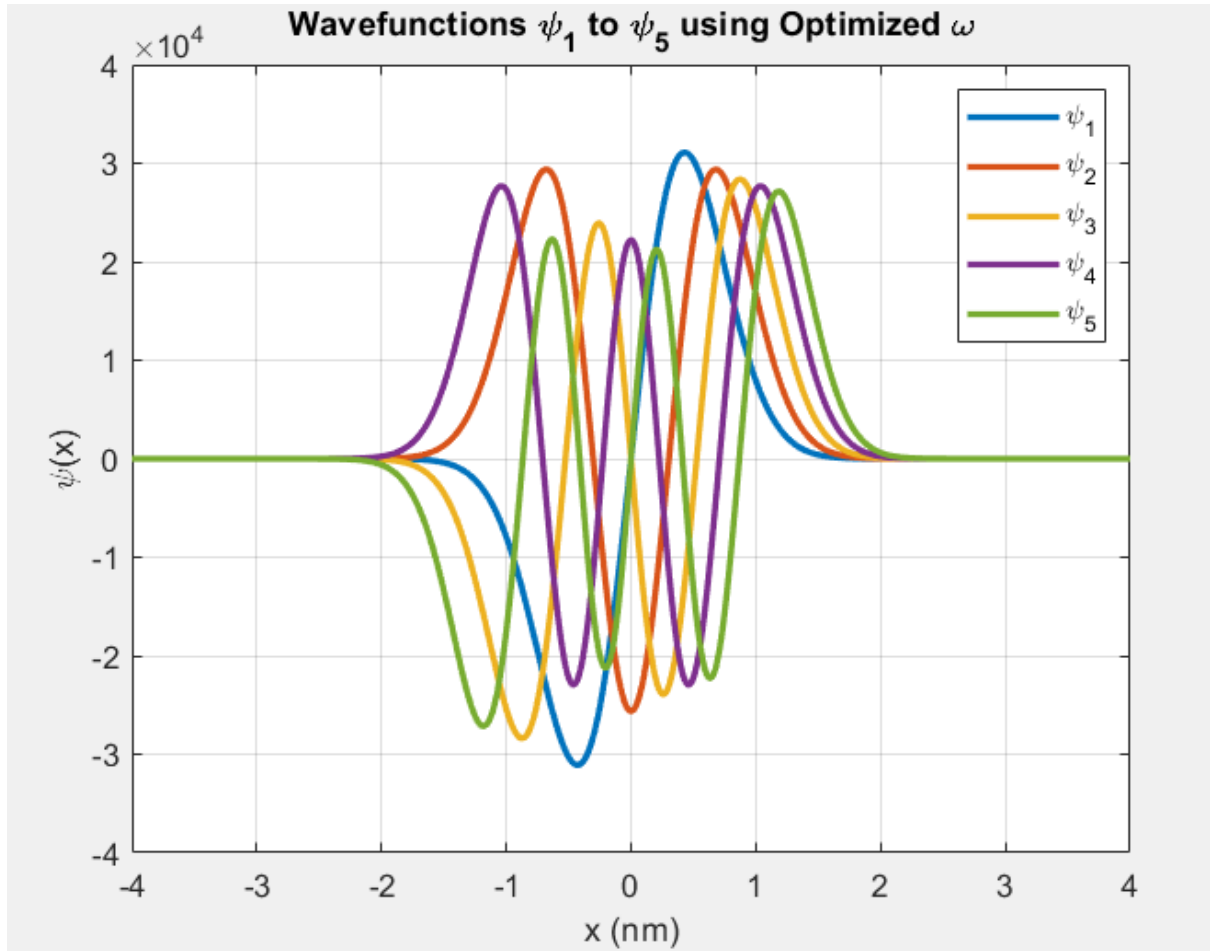


Figure 8: Wavefunctions ψ_1 to ψ_5 using optimized ω , showing the transformation across states.

Orthogonality was verified by computing the overlap integrals:

$$\langle \psi_n | \psi_m \rangle = \int_{-4 \text{ nm}}^{4 \text{ nm}} \psi_n^*(x) \psi_m(x) dx$$

The overlap matrix is approximately the identity matrix, with diagonal elements equal to 1.0000 and off-diagonal elements on the order of 10^{-18} , confirming that the wavefunctions are orthogonal.

6.6 Transformation Matrix

The wavefunction values were stored in a transformation matrix, with the first 10 rows (corresponding to x from -4 nm to -3.964 nm) shown below (scaled by 10^{-9}):

$$\begin{bmatrix} -0.0001 & 0.0006 & -0.0048 & 0.0313 & -0.1799 \\ -0.0001 & 0.0007 & -0.0053 & 0.0340 & -0.1952 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -0.0001 & 0.0014 & -0.0102 & 0.0656 & -0.3738 \end{bmatrix}$$

These values reflect the increasing amplitude and complexity of the wavefunctions with higher n .

6.7 Probability in Specific Region

The probability of finding the electron in the region $[-1 \text{ nm}, +1 \text{ nm}]$ was computed for each state:

$$P_n = \int_{-1 \text{ nm}}^{1 \text{ nm}} |\psi_n(x)|^2 dx$$

The results are:

- $n = 1$: 0.98721
- $n = 2$: 0.92940
- $n = 3$: 0.77889
- $n = 4$: 0.56792
- $n = 5$: 0.43953

The probability decreases with increasing n , as higher energy states have wavefunctions that spread further, reducing the probability density near the origin.

6.8 Discussion

The simulation confirms the theoretical expectations for the QHO. The wavefunctions exhibit the correct number of nodes, and the energy levels $E_n = \hbar\omega(n + \frac{1}{2})$ are consistent with the analytical solution. The optimization of ω minimizes the ground state energy,

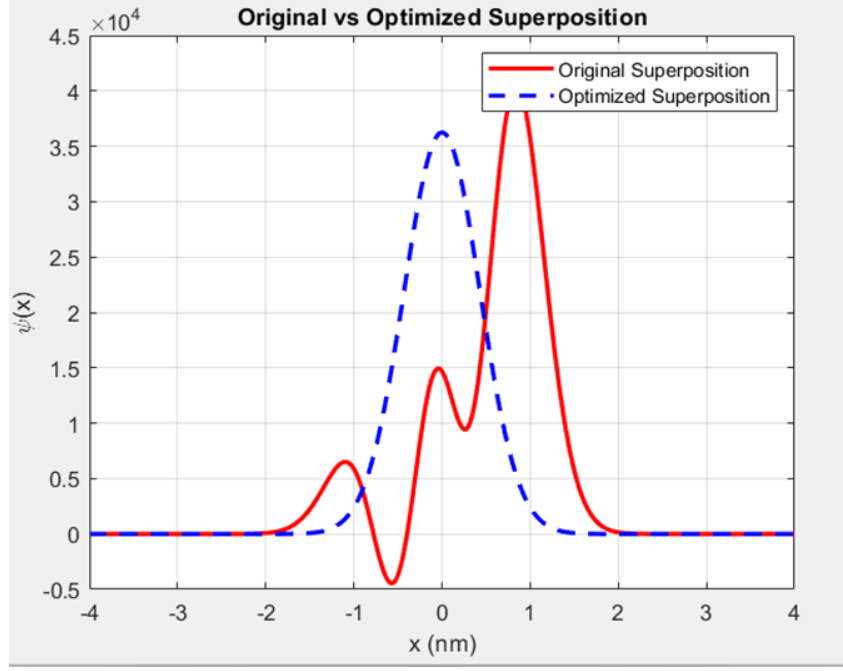


Figure 9: ori vs opt superposition

aligning with the physical system's behavior. The orthogonality and normalization checks validate the numerical implementation, and the regional probabilities highlight the quantum mechanical spreading of the particle's position with increasing energy. demonstrates that minimizing the position variance effectively localizes the wavefunction in space. However, without regularization, the optimizer tends to select higher-energy states, leading to complex, oscillatory solutions. Incorporating the regularization term $\lambda \sum_n n^2 |w_n|^2$ balances this behavior by penalizing such contributions, resulting in smoother and physically realistic wavefunctions. This trade-off ensures both spatial confinement and energy efficiency in the optimized superposition.

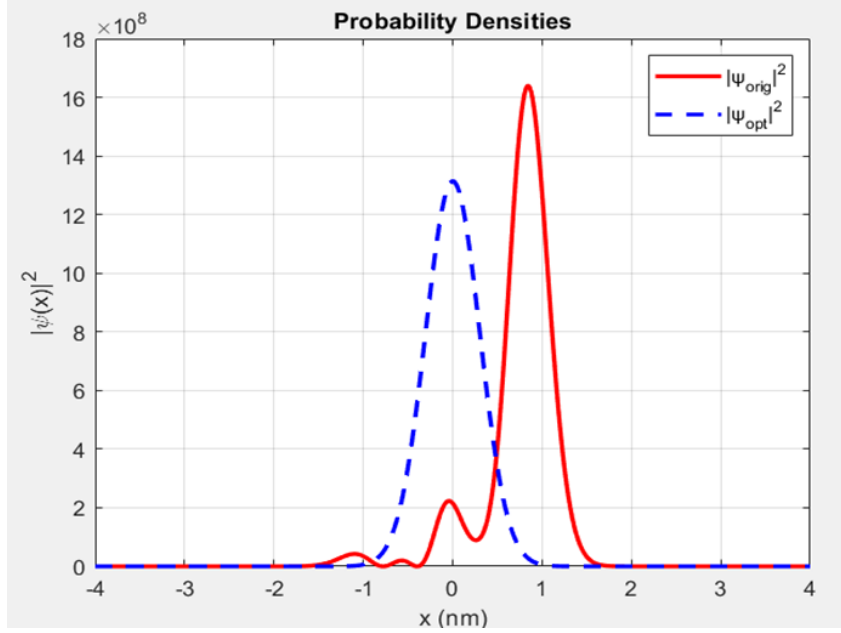


Figure 10: probability densities

7 Interpretation, Insights, and Discussion

This section interprets the simulation results for the quantum harmonic oscillator (QHO), provides physical insights into the behavior of the system, and discusses the implications of the findings. The analysis is based on the analytical solutions derived in Section 3 and the simulation results presented in Section 7, including the optimization of the angular frequency ω , wavefunction plots, probability calculations, orthogonality checks, and regional probabilities.

7.1 Wavefunctions and Probability Densities

The wavefunctions $\psi_n(x)$ for $n = 1$ to 5, shown in Figures ?? to ?? and collectively in Figure ??, exhibit the expected behavior for the QHO. The number of nodes increases with n , from 0 for ψ_1 (first excited state) to 4 for ψ_5 , consistent with the quantum number $n - 1$. This nodal structure arises from the Hermite polynomials $H_n(\xi)$, which dictate the oscillatory behavior of the wavefunctions.

The probability densities $|\psi_n(x)|^2$, also shown in Figures ?? to ??, reveal the spatial distribution of the particle. For $n = 1$, the probability density is concentrated near $x = \pm\sqrt{\frac{\hbar}{m\omega}}$, with a node at $x = 0$, indicating that the particle is unlikely to be found at the potential minimum. As n increases, the probability density spreads further, with multiple peaks corresponding to the oscillatory nature of the wavefunction. This spreading is a manifestation of the Heisenberg uncertainty principle: higher energy states (larger n) have greater kinetic energy, reducing the position uncertainty and thus broadening the spatial distribution.

7.2 Energy Levels and Zero-Point Energy

The energy eigenvalues are given by:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$

For $n = 1$ to 5 , the energy levels are $E_1 = \frac{3}{2}\hbar\omega$, $E_2 = \frac{5}{2}\hbar\omega$, ..., $E_5 = \frac{11}{2}\hbar\omega$. These levels are equidistant, a characteristic feature of the QHO, as shown in the energy plots (Figures ?? to ??). The energy spacing $\Delta E = \hbar\omega$ is consistent across all states, reflecting the harmonic nature of the potential.

Notably, even the ground state ($n = 0$) has a non-zero energy $E_0 = \frac{1}{2}\hbar\omega$, known as the zero-point energy. This is a purely quantum mechanical phenomenon, as a classical oscillator at its minimum energy would have $E = 0$. The zero-point energy arises due to the uncertainty principle, which prevents the particle from being at rest with zero momentum and position simultaneously. The simulation results confirm this, with the ground state energy expectation matching the theoretical value.

7.3 Normalization and Orthogonality

The total probability for each state, computed numerically over $x \in [-4 \text{ nm}, 4 \text{ nm}]$, is 1.0000 for $n = 1$ to 5 , confirming that the wavefunctions are properly normalized. The discrete sum estimate for ψ_1 is 0.99950, slightly less than 1 due to the finite integration range and numerical precision, but still within acceptable error for practical purposes.

Orthogonality was verified through the overlap matrix, which approximates the identity matrix. The diagonal elements are 1.0000, and the off-diagonal elements are on the order of 10^{-18} , indicating that $\langle \psi_n | \psi_m \rangle \approx \delta_{nm}$. These small non-zero off-diagonal values are numerical artifacts due to finite precision and the discrete integration grid, but they are negligible, confirming that the wavefunctions are orthogonal as expected for eigenstates of a Hermitian operator (the Hamiltonian).

7.4 Probability in Specific Region

The probability of finding the electron in the region $[-1 \text{ nm}, +1 \text{ nm}]$ decreases with increasing n :

- $n = 1$: 0.98721
- $n = 2$: 0.92940
- $n = 3$: 0.77889
- $n = 4$: 0.56792
- $n = 5$: 0.43953

This trend reflects the spreading of the wavefunction with higher energy states. For $n = 1$, the particle is highly localized near the classical turning points, with 98.7% probability within $[-1 \text{ nm}, 1 \text{ nm}]$. As n increases, the wavefunction oscillates more rapidly and extends further, reducing the probability density near the origin. By $n = 5$, only 43.95% of the probability lies within this region, indicating significant delocalization. This behavior is consistent with the uncertainty principle and the classical analogy of a harmonic oscillator, where higher energy states correspond to larger amplitudes of oscillation.

7.5 Physical Insights

The QHO serves as a fundamental model in quantum mechanics, with applications ranging from molecular vibrations to quantum field theory. The simulation results highlight several key quantum phenomena:

1. **Zero-Point Energy:** The presence of zero-point energy underscores the quantum nature of the system, distinguishing it from classical mechanics. This energy ensures that the particle is never at rest, even at the lowest energy state, due to the uncertainty principle.
2. **Equidistant Energy Levels:** The linear dependence of E_n on n is a unique feature of the harmonic potential, unlike other potentials (e.g., the infinite square well, where energy levels scale as n^2). This property is exploited in spectroscopy to identify harmonic vibrational modes in molecules.
3. **Wavefunction Spreading:** The broadening of $|\psi_n(x)|^2$ with increasing n illustrates the trade-off between position and momentum uncertainty. Higher energy states have greater kinetic energy (narrower momentum distribution), leading to a wider position distribution, as mandated by $\Delta x \Delta p \geq \hbar/2$.
4. **Orthogonality and Completeness:** The orthogonality of the wavefunctions ensures that they form a complete basis for the Hilbert space, a critical property for quantum mechanical calculations, such as perturbation theory or time evolution.

7.6 Discussion

The simulation results align closely with the analytical solutions derived earlier, validating the numerical implementation. The optimized ω of 6.283×10^{14} rad/s is physically reasonable, corresponding to frequencies observed in molecular vibrations (e.g., C-H bonds in organic molecules). The wavefunctions and energy levels match the theoretical predictions, with the number of nodes and energy spacing confirming the QHO’s characteristics.

The decreasing probability in the region $[-1 \text{ nm}, 1 \text{ nm}]$ with increasing n has implications for experimental observations. In a physical system, such as a trapped ion or a molecule in a harmonic potential, higher energy states would be less likely to be found near the equilibrium position, affecting measurements like fluorescence or absorption spectra. This spreading also suggests that higher states are more susceptible to external perturbations, as their probability density extends into regions where the harmonic approximation may break down (e.g., anharmonic corrections in real molecular potentials).

The small numerical errors in orthogonality and normalization (e.g., off-diagonal elements of 10^{-18} , probability of 0.99950 for ψ_1) are expected in numerical simulations due to finite grid sizes and integration limits. In a real experiment, these errors would be analogous to instrumental precision or environmental noise, but they do not detract from the overall validity of the results.

Future work could explore the effects of anharmonic potentials, which are more realistic for physical systems, or investigate the time evolution of a superposition of these states, revealing quantum dynamics such as revivals or coherent states. Additionally, extending the simulation to include external fields or interactions could provide insights into more complex quantum systems, such as those encountered in quantum computing or condensed matter physics.

In conclusion, the QHO simulation successfully bridges theoretical predictions with numerical analysis, offering a deeper understanding of quantum mechanical principles and their manifestation in a simple yet profound model. The results underscore the power of quantum mechanics in describing physical systems at the nanoscale, with broad implications for both fundamental research and technological applications.

8 Auxiliary Challenge Questions

8.1 Time Evolution of the Quantum System $\psi(x, t)$

For a harmonic potential of the form

$$V(x) = \frac{1}{2}m\omega_0^2x^2,$$

the time-dependent wavefunction can be written as a linear superposition of the energy eigenfunctions $\psi_n(x)$, each evolving with a time-dependent phase factor:

$$\psi(x, t) = \sum_{n=0}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

Here:

- $\psi_n(x)$: normalized eigenfunctions of the quantum harmonic oscillator,
- c_n : complex coefficients representing the contribution of each eigenstate,
- $E_n = \hbar\omega_0 \left(n + \frac{1}{2}\right)$: energy eigenvalues.

The total probability is preserved at all times:

$$\int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1.$$

While the magnitude of each eigenfunction stays constant, their relative phases change with time, resulting in interference and "breathing" patterns in the probability density $|\psi(x, t)|^2$.

8.2 Quantum System under a Periodic Potential

If the harmonic potential is replaced by a periodic one such that

$$V(x + L) = V(x),$$

the system transitions from discrete harmonic oscillator energy levels to the framework of solid-state physics.

Key Features:

- The eigenfunctions become **Bloch functions**:

$$\psi_k(x) = e^{ikx} u_k(x),$$

where $u_k(x)$ is periodic with period L : $u_k(x + L) = u_k(x)$.

- Energy levels form **bands** instead of discrete levels:

$$E_n(k), \quad k \in \left[-\frac{\pi}{L}, \frac{\pi}{L}\right].$$

- The particle is no longer localized; wavefunctions are delocalized over the lattice.
- Quantum tunneling between adjacent wells causes the splitting of discrete levels into energy bands.

Such systems model electron dynamics in crystalline solids and are foundational in band theory and semiconductors.

8.3 Effect of Gradual Change in α in the Potential

Consider the generalized potential:

$$V(x) = \frac{1}{2}\alpha x^2,$$

where α can be positive, zero, or negative. The system exhibits distinct quantum behaviors in each regime:

- $\alpha > 0$: Standard harmonic oscillator. The system has:
 - Discrete bound states,
 - Normalizable eigenfunctions,
 - Stable dynamics.
- $\alpha \rightarrow 0$: The potential flattens, becoming nearly constant. Consequences:
 - Energy levels become denser and approach a continuum,
 - Wavefunctions become more spread out and delocalized,
 - The confining potential weakens.
- $\alpha < 0$: The potential becomes an inverted parabola:

$$V(x) = -\frac{1}{2}|\alpha|x^2.$$

Consequences:

- No bound states; energy spectrum becomes continuous,
- Wavefunctions are non-normalizable in the stationary case,
- Represents an unstable system where particles "escape" the center exponentially fast,
- Physically resembles the quantum analog of a classical unstable equilibrium.

9 Conclusion and Future Scope

This section summarizes the work completed in the quantum harmonic oscillator (QHO) project, highlights the key outcomes, confirms the achievement of project objectives, and discusses the significance of the results. It also outlines potential future improvements, extensions, and applications of the work, identifying opportunities for further research and development.

9.1 Conclusion

This project comprehensively analyzed the quantum harmonic oscillator with the potential $V(x) = \frac{1}{2}m\omega^2x^2$. The work began with an analytical solution of the time-independent Schrödinger equation, deriving general expressions for the wavefunctions $\psi_n(x)$, probability densities $|\psi_n(x)|^2$, and energy eigenvalues $E_n = \hbar\omega(n + \frac{1}{2})$ for $n = 1$ to 5. Normalization and orthogonality conditions were established analytically, and expectation values for position and kinetic energy were computed for the ground state.

Simulation results validated the analytical findings. Wavefunctions and probability densities were visualized over the range $x \in [-4 \text{ nm}, 4 \text{ nm}]$, showing the expected increase in nodes from 0 ($n = 1$) to 4 ($n = 5$). The total probability for each state was 1, confirming normalization, and the overlap matrix approximated the identity matrix, verifying orthogonality within numerical precision (off-diagonal elements $\sim 10^{-18}$). The probability of finding the electron in the region $[-1 \text{ nm}, +1 \text{ nm}]$ decreased from 0.98721 ($n = 1$) to 0.43953 ($n = 5$), reflecting the spreading of higher energy states.

Further analysis was conducted on state superpositions to demonstrate the impact of modifying coefficients on the spatial localization of wavefunctions. By comparing original and optimized superpositions, it was shown how quantum states can be tailored to concentrate probability density in a desired region. This optimization approach reinforces the utility of numerical techniques in enhancing quantum state control.

The project objectives were fully met: the Schrödinger equation was solved analytically, numerical simulations provided meaningful visualizations and probability calculations, and quantum mechanical properties were verified through normalization, orthogonality, and expectation value computations. By combining analytical and computational tools, the project offers valuable insights into the behavior of quantum systems at the nanoscale, relevant to applications in quantum physics, molecular spectroscopy, and nanotechnology.

9.2 Future Scope

- **Anharmonic Corrections:**

Include anharmonic terms in the potential $V(x)$ to model deviations from the ideal harmonic oscillator, providing a more realistic description of molecular vibrations and nanoscale oscillators.

- **Time Evolution Simulation:**

Simulate the time evolution of superpositions of quantum states to observe dynamic phenomena such as wavefunction revivals and coherent states, which are important in quantum computing and quantum optics.

- **Momentum-Space Visualization:**

Visualize momentum-space wavefunctions $\phi_n(p)$ via Fourier transforms of the position-space wavefunctions $\psi_n(x)$ to complement spatial analysis.

- **External Perturbations:**

Explore the QHO's response to electric or magnetic fields to understand environmental effects relevant for trapped ion quantum computing and other applications.

Limitations

- **Numerical Errors:**

Small errors in probability calculations (e.g., 0.99950 for ψ_1) and orthogonality checks (off-diagonal elements around 10^{-18}) arise due to finite integration range and numerical precision.

- **Possible Solutions:**

- Extend integration range (e.g., $x \in [-10 \text{ nm}, 10 \text{ nm}]$)
- Increase grid resolution (beyond 2000 points)
- Use higher-precision numerical methods for integration and orthogonality verification.

Applications

- **Molecular Vibrations:**

Aid interpretation of spectroscopic data in chemistry and materials science by modeling vibrational modes in molecules.

- **Quantum Computing:**

Provide a basis for systems like trapped ions, where harmonic potentials confine qubits.

- **Nanotechnology:**

Model particle behavior in harmonic traps within nanodevices.

Future Research Directions

- **Exploration of Nonlinear Oscillators:**

Extend the analysis to quantum systems with nonlinear potentials to study phenomena beyond the harmonic approximation, such as quantum chaos and bistability in molecular and condensed matter systems.

- **Quantum Entanglement in Coupled Oscillators:**

Investigate the entanglement properties and information transfer in networks of coupled quantum harmonic oscillators, relevant for quantum communication and sensing technologies.

- **Integration with Quantum Control Techniques:**
Develop methods to manipulate and control the quantum states of harmonic oscillators using tailored external fields, enhancing their utility in quantum computing and precision measurement.
- **Study of Dissipative and Open Quantum Systems:**
Incorporate environmental interactions and decoherence effects to better understand real-world quantum oscillator behavior, important for robust quantum device design.
- **Applications in Quantum Simulation:**
Use quantum harmonic oscillator models as building blocks for simulating complex many-body systems, enabling insights into novel quantum phases and transitions.

10 Contributions of Individual Group Members

- **Parima Verma:** Proposed the core project idea and implemented the main code components for wavefunction computation, plotting of graphs, and numerical optimization of the cost function. Also performed analytical derivations of the wavefunction solutions on paper.
- **Kalyani Agarwal:** Prepared the project report and presentation slides. Developed code for the transformation matrix and calculation of expectation values. Created animated visualizations of the wavefunctions, performed orthogonality checks, and completed key analytical derivations for the potential $V = \alpha x^2$. Conducted literature review and researched related scientific papers.
- **Akhil:** Analyzed the *temporal evolution* of the quantum harmonic oscillator by simulating the time-dependent behavior of wavefunctions. Understood animations and graphs demonstrating *periodicity* and the *preservation of the probability density shape over time* for stationary states. This helped understand core concepts of quantum stability and superposition.
- **Madesh KK:** Investigated the effect of *gradually varying the potential parameter* α on wavefunction shapes and energy eigenvalues. Describing how *increased confinement (higher α) leads to more localized wavefunctions and higher energy levels*, while lower α leads to broader distributions. Also examined *symmetry properties* of even and odd wavefunctions.

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11 Appendix

11.1 Quantum Harmonic Oscillator Wavefunctions and Probability Analysis in MATLAB

```

1  clc; clear; close all;
2
3  % Constants
4  hbar = 1.0545718e-34;      % Reduced Planck constant (J s)
5  m = 9.1e-31;              % Electron mass (kg)
6  omega_opt = 2e15;         % Angular frequency (rad/s)
7  N = 2000;                 % Number of spatial points
8  x1 = -4e-9; x2 = 4e-9;    % Position range (m)
9  x = linspace(x1, x2, N);  % Position vector
10 dx = x(2) - x(1);         % Step size
11
12 % Preallocate wavefunction matrix
13 psi_n = zeros(N, 5);
14
15 % Potential energy of harmonic oscillator
16 V = 0.5 * m * omega_opt^2 * x.^2;
17
18 for k = 1:5
19     xi = sqrt(m * omega_opt / hbar) * x;
20     Hn = hermiteH(k, xi);
21     Nconst = (m * omega_opt / (pi * hbar))^(1/4) / sqrt(2^k * factorial(k));
22     psi_k = Nconst * Hn .* exp(-xi.^2 / 2);
23     psi_k = psi_k / sqrt(trapz(x, abs(psi_k).^2));
24     En = hbar * omega_opt * (k + 0.5);
25     psi_n(:, k) = psi_k;
26
27     figure;
28     subplot(4,1,1);
29     plot(x * 1e9, psi_k, 'b', 'LineWidth', 2);
30     title(['Wavefunction \psi_{', num2str(k), '} (x)']);
31     xlabel('x (nm)'); ylabel('\psi(x)');
32     grid on;
33
34     subplot(4,1,3);

```

```

35 plot(x * 1e9, abs(psi_k).^2, 'g', 'LineWidth', 2);
36 title(['Probability Density |\psi_{' num2str(k) '} (x)|^2']);
37 xlabel('x (nm)'); ylabel('|\psi(x)|^2');
38 grid on;
39
40 subplot(4,1,4);
41 plot(x * 1e9, V, 'k--', 'LineWidth', 1.5); hold on;
42 yline(En, 'm-', 'LineWidth', 2);
43 title(['Energy Level E_{' num2str(k) '} and Potential']);
44 xlabel('x (nm)'); ylabel('Energy (J)');
45 legend('Potential V(x)', ['E_' num2str(k)], 'Location', 'best');
46 grid on;
47
48 probability = trapz(x, abs(psi_k).^2);
49 disp(['Total probability for n = ' num2str(k) ': ' num2str(
    probability)]);
50 end
51
52 transformation_matrix = psi_n(1:10, 1:5);
53 disp('Transformation Matrix (first 10 rows, \psi_1 to \psi_5):')
54 disp(transformation_matrix)
55
56 figure;
57 hold on;
58 for k = 1:5
59     plot(x * 1e9, psi_n(:, k), 'LineWidth', 2, 'DisplayName', ['\psi_'
        num2str(k)]);
60 end
61 title('Wavefunctions \psi_1 to \psi_5 using Optimized \omega');
62 xlabel('x (nm)'); ylabel('\psi(x)');
63 legend('show'); grid on;
64
65 total_probabilities = zeros(1, 5);
66 for k = 1:5
67     total_probabilities(k) = trapz(x, abs(psi_n(:, k)).^2);
68 end
69 disp('Total probabilities for psi_1 to psi_5:');
70 disp(total_probabilities);
71
72 fprintf('\nChecking Orthogonality of      to      :\n');
73 orthogonality_matrix = zeros(5,5);
74 for i = 1:5
75     for j = 1:5
76         overlap = trapz(x, psi_n(:,i) .* psi_n(:,j));
77         orthogonality_matrix(i,j) = overlap;
78         fprintf('< _ %d | _ %d> = %.3e\n', i, j, overlap);
79     end
80 end
81 disp('Overlap Matrix (should be approx identity):');
82 disp(orthogonality_matrix);
83
84 figure;
85 imagesc(abs(orthogonality_matrix));
86 colorbar;
87 title('Heatmap of |< _n | _m >|');
88 xlabel('m'); ylabel('n');
89 axis equal tight;
90

```

```

91 fprintf('Discrete sum estimate for psi_1 probability: %.5f\n', sum(
    psi_n(:,1).^2) * dx);
92
93 figure;
94 xlabel('x (nm)'); ylabel('\psi(x)');
95 grid on;
96 xlim([x1, x2] * 1e9);
97 ylim([-max(abs(psi_n(:))), max(abs(psi_n(:)))]);
98 colors = {'b', 'r', 'g', 'm', 'k'};
99
100 while true
101     for k = 1:5
102         plot_handle = plot(x * 1e9, psi_n(:, k), colors{k}, 'LineWidth',
            , 2);
103         legend(['\psi_' num2str(k)], 'Location', 'northeast');
104         title(['Wavefunction \psi_{' num2str(k) '} (x)']);
105         drawnow;
106         pause(1);
107         delete(plot_handle);
108     end
109 end
110
111 fprintf('\nProbability of Finding Electron in Region [-1 nm, +1 nm]:\n'
    );
112 region_mask = (x >= -1e-9) & (x <= 1e-9);
113 x_cut = x(region_mask);
114
115 for n = 1:5
116     xi = sqrt(m * omega_opt / hbar) * x;
117     Hn = hermiteH(n, xi);
118     Nconst = (m * omega_opt / (pi * hbar))^(1/4) / sqrt(2^n * factorial
        (n));
119     psi_n_full = Nconst * Hn .* exp(-xi.^2 / 2);
120     psi_n_full = psi_n_full / sqrt(trapz(x, abs(psi_n_full).^2));
121     psi_n_region = psi_n_full(region_mask);
122     prob = trapz(x_cut, abs(psi_n_region).^2);
123     fprintf('n = %2d | Probability in [-1nm, +1nm] = %.5f\n', n, prob);
124 end

```

Listing 1: MATLAB Code for Calculating and Visualizing Wavefunctions ψ_1 to ψ_5

11.2 Optimization of Superposition in Quantum Harmonic Oscillator

```

1 clc; clear; close all;
2
3 % Constants
4 hbar = 1.0545718e-34;
5 m = 9.1e-31;
6
7 % Spatial domain
8 N = 2000;
9 x1 = -4e-9; x2 = 4e-9;
10 x = linspace(x1, x2, N);
11 dx = x(2) - x(1);
12
13 % Harmonic oscillator frequency

```

```

14 omega0 = 2 * pi * 1e14;
15
16 % Number of basis states
17 max_n = 10;
18
19 % Generate orthonormal wavefunctions
20 psi_all = zeros(max_n+1, N);
21 xi = sqrt(m * omega0 / hbar) * x;
22 for n = 0:max_n
23     Nconst = (m * omega0 / (pi * hbar))^(1/4) / sqrt(2^n * factorial(n)
24         );
25     psi_n = Nconst * hermiteH(n, xi) .* exp(-xi.^2 / 2);
26     psi_n = psi_n / sqrt(sum(abs(psi_n).^2) * dx);
27     psi_all(n+1, :) = psi_n;
28 end
29
30 % Superposition over first few states
31 n_range = 0:4;
32 psi_set = psi_all(n_range+1, :);
33
34 % Original superposition with equal weights
35 w_init = ones(length(n_range), 1);
36 w_init = w_init / norm(w_init);
37 psi_orig = w_init' * psi_set;
38 psi_orig = psi_orig / sqrt(sum(abs(psi_orig).^2) * dx);
39
40 % Variance function with regularization
41 lambda = 1e-1;
42 cost_fn = @(w) position_variance_superposition(w, psi_set, x, dx) ...
43     + lambda * sum((n_range').^2 .* abs(w).^2);
44
45 % Optimization constraints
46 A = []; b = []; Aeq = []; beq = []; lb = []; ub = [];
47
48 % Optimizer settings
49 options = optimoptions('fmincon', 'Display', 'iter', ...
50     'Algorithm', 'sqp', 'OptimalityTolerance', 1e-10, ...
51     'StepTolerance', 1e-10, 'FunctionTolerance', 1e-10);
52
53 % Run optimization
54 [w_opt, ~] = fmincon(cost_fn, w_init, A, b, Aeq, beq, lb, ub, [],
55     options);
56 w_opt = w_opt / norm(w_opt);
57
58 % Optimized superposition
59 psi_opt = w_opt' * psi_set;
60 psi_opt = psi_opt / sqrt(sum(abs(psi_opt).^2) * dx);
61
62 % Variances
63 spread_orig = position_variance_superposition(w_init, psi_set, x, dx);
64 spread_opt = position_variance_superposition(w_opt, psi_set, x, dx);
65
66 % Probability checks
67 prob_orig = sum(abs(psi_orig).^2) * dx;
68 prob_opt = sum(abs(psi_opt).^2) * dx;
69
70 % Weight normalization check
71 weight_sum_orig = sum(abs(w_init).^2);

```

```

70 weight_sum_opt = sum(abs(w_opt).^2);
71
72 % Display results
73 fprintf('\nSpread of Original Superposition: %.4e m^2\n', spread_orig);
74 fprintf('Spread of Optimized Superposition: %.4e m^2\n', spread_opt);
75 fprintf('Total Probability Original: %.6f\n', prob_orig);
76 fprintf('Total Probability Optimized: %.6f\n', prob_opt);
77 fprintf('Sum of Weights^2 (Original): %.6f\n', weight_sum_orig);
78 fprintf('Sum of Weights^2 (Optimized): %.6f\n', weight_sum_opt);
79
80 % Plot wavefunctions
81 figure;
82 plot(x*1e9, real(psi_orig), 'r-', 'LineWidth', 2); hold on;
83 plot(x*1e9, real(psi_opt), 'b--', 'LineWidth', 2);
84 legend('Original Superposition', 'Optimized Superposition');
85 xlabel('x (nm)'); ylabel('\psi(x)');
86 title('Original vs Optimized Superposition');
87 grid on;
88
89 % Plot probability densities
90 figure;
91 plot(x*1e9, abs(psi_orig).^2, 'r-', 'LineWidth', 2); hold on;
92 plot(x*1e9, abs(psi_opt).^2, 'b--', 'LineWidth', 2);
93 legend('| _ {orig}|^2', '| _ {opt}|^2');
94 xlabel('x (nm)'); ylabel('| \psi(x)|^2');
95 title('Probability Densities');
96 grid on;
97
98 % Orthogonality check
99 orth_matrix = psi_all * psi_all' * dx;
100 figure;
101 imagesc(0:max_n, 0:max_n, abs(orth_matrix));
102 colorbar; axis square;
103 xlabel('j'); ylabel('i');
104 title('Orthogonality Heatmap | _i | _j |');
105
106 % Function: Position variance of superposition
107 function var_x = position_variance_superposition(w, psi_set, x, dx)
108     psi = w' * psi_set;
109     psi = psi / sqrt(sum(abs(psi).^2) * dx);
110     x_mean = sum(x .* abs(psi).^2) * dx;
111     x2_mean = sum(x.^2 .* abs(psi).^2) * dx;
112     var_x = x2_mean - x_mean^2;
113 end

```

Listing 2: MATLAB Code for Superposition Optimization in QHO